VR INJECTION ADMINISTRATION

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Team IDK

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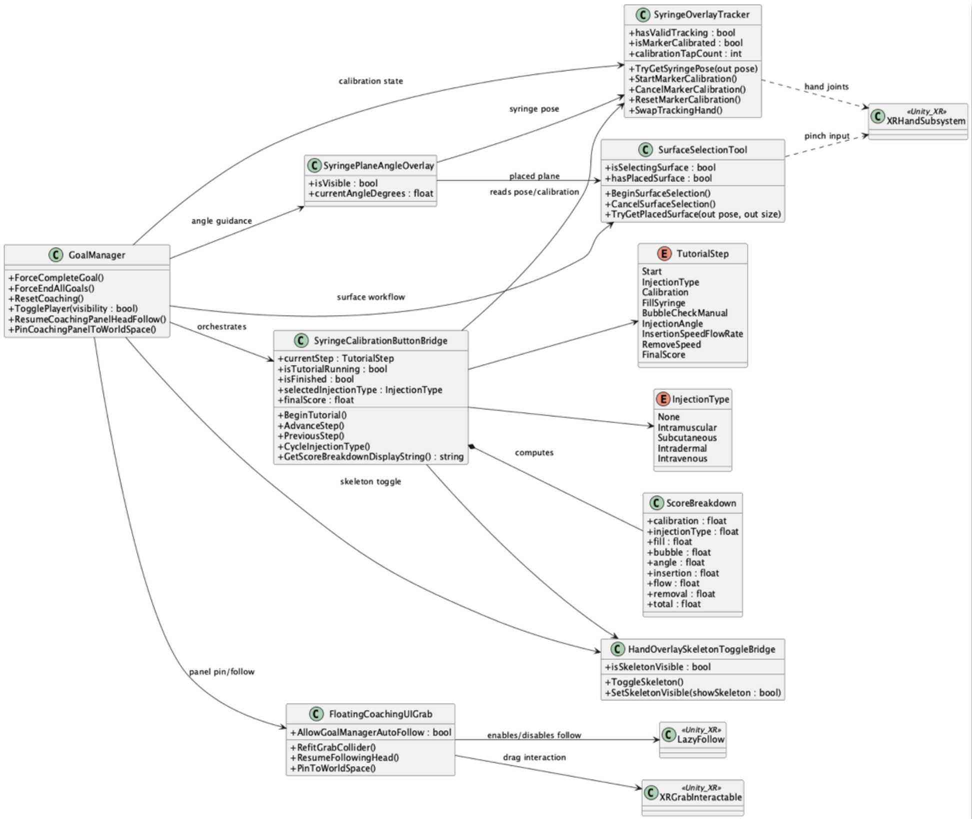
OUR IDEA

We build a system that doctors and nurses in training can use to practice administering injections.

Initially the plan was to create a more general-purpose system that can track real world objects

"As we are DASS TAs, lets have the prerequisite UML diagram"

- DIVIJH



SyringeOverlayTracker
+hasValidTracking : bool
+isMarkerCalibrated : bool
+calibrationTapCount : int
+TryGetSyringePose(out pose)
+StartMarkerCalibration()
+CancelMarkerCalibration()
+ResetMarkerCalibration()
+SwapTrackingHand()
hand joints
«Unity_XR»
XRHandSubsystem
pinch input
calibration state
syringe pose
SyringePlaneAngleOverlay
+isVisible : bool
+currentAngleDegrees : float
placed plane
reads pose/calibration
SurfaceSelectionTool
+isSelectingSurface : bool
+hasPlacedSurface : bool
+BeginSurfaceSelection()
+CancelSurfaceSelection()
+TryGetPlacedSurface(out pose, out size)
angle guidance
GoalManager
+ForceCompleteGoal()
+ForceEndAllGoals()
+ResetCoaching()
+TogglePlayer(visibility : bool)
+ResumeCoachingPanelHeadFollow()
+PinCoachingPanelToWorldSpace()
surface workflow
TutorialStep
Start
InjectionType
Calibration
FillSyringe
BubbleCheckManual
InjectionAngle
InsertionSpeedFlowRate
RemoveSpeed
FinalScore
orchestrates
SyringeCalibrationButtonBridge
+currentStep : TutorialStep
+isTutorialRunning : bool
+isFinished : bool
+selectedInjectionType : InjectionType
+finalScore : float
+BeginTutorial()
+AdvanceStep()
+PreviousStep()
+CycleInjectionType()
+GetScoreBreakdownDisplayString() : string
InjectionType
None
Intramuscular
Subcutaneous
Intradermal
Intravenous
computes
skeleton toggle
ScoreBreakdown
+calibration : float
+injectionType : float
+fill : float
+bubble : float
+angle : float
+insertion : float
+flow : float
+removal : float
+total : float
HandOverlaySkeletonToggleBridge
+isSkeletonVisible : bool
+ToggleSkeleton()
+SetSkeletonVisible(showSkeleton : bool)
panel pin/follow
FloatingCoachingUIGrab
+AllowGoalManagerAutoFollow : bool
+RefitGrabCollider()
+ResumeFollowingHead()
+PinToWorldSpace()
enables/disables follow
«Unity_XR»
LazyFollow
drag interaction
«Unity_XR»
XRGrabInteractable

01.

ABOUT US

We are a team of 2 UG 3 student under Prof. Raghu

ABOUT THE SYSTEM

Built using Unity
Deployed on the Meta Quest 3
Uses hand tracking

OUR GOALS

NO EXTERNAL CAMERAS

No external camera need to be present

PERFORMANCE

The system needs to run on consumer hardware realtime

ACCURACY

The system need to be able to track the syringe accurately

02.

IMPLEMENTATION

IMPLEMENTATION

DUAL-TRACKING ARCHITECTURE:

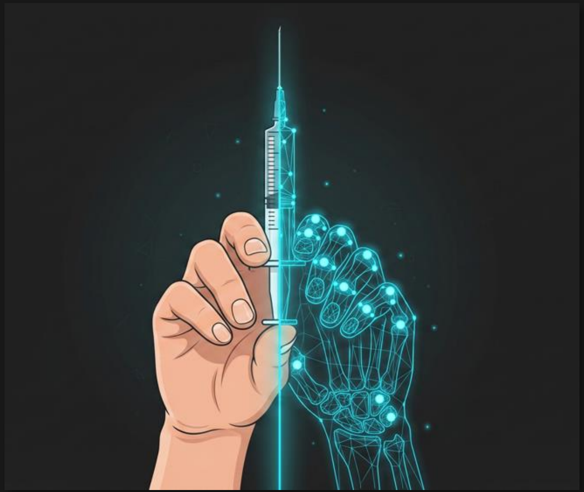
- Marker-Based Syringe Tracking: High-precision pose estimation of the syringe tool.
- XR Hands Integration: Real-time skeleton overlays for ergonomic hand placement feedback.

REAL-TIME METRIC CALCULATION:

- Velocity Estimation: Frame-by-frame derivation of needle tip speed to calculate Insertion Speed and Removal Speed in cm/sec.
- Entry Angle Logic: Using Vector3.Angle against the gravity/surface normal to provide dynamic angle guidance.
- Flow Rate Simulation: A mathematical derivation based on insertion depth and plunger movement (forwardSpeed * 0.15f).

COMPREHENSIVE SCORING ENGINE:

- Evaluation across 8 distinct categories (Calibration, Angle, Flow, etc.), resulting in a ScoreBreakdown stored as a structured data object.



CHALLENGES

PRECISION

- Aligning a virtual needle model with physical syringes in MR space while accounting for tracker jitter.

MARKER OCCLUSION

- Managing ”dead zones” in marker tracking where the syringe body blocks the headset sensors during.

CONTEXTUAL UX DESIGN

- Balancing the information density of the Coaching Panel (metrics like angle and speed) without cluttering the user’s field of view.

MR ENVIRONMENTAL FACTORS

- Ensuring the Surface and Marker Selection Tool works reliably across various real-world textures and lighting via AR Foundation/Meta Passthrough.

03.

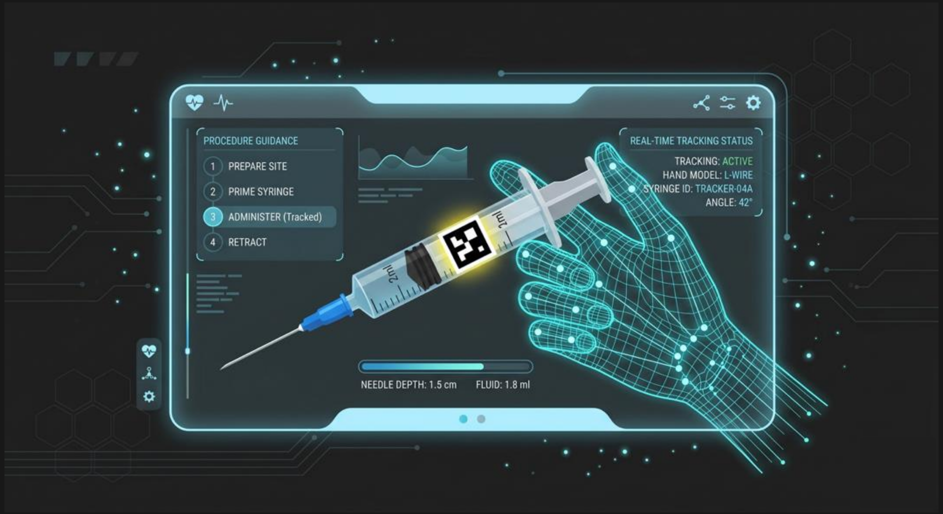
DEMO

DEMO

Demo here

04.

FUTURE WORK



FUTURE WORK

UI, UI, UI !!

We plan to integrate a more robust marker-based system to further improve the accuracy.

Additionally, we also plan to add more grip profiles to improve the tracking accuracy

THANKS!

QUESTIONS?